

QUESTIONS FOR CHAPTER 6 — DEVELOPMENT OF WORK PLAN

1. As discussed in this chapter, the **work environment** of a **project manager** can have a **significant impact** on his or her ability to manage projects. Describe factors that may help or hinder the work of a project manager in each of the following **organizational structures**: **functional organization**, **discipline organization**, and **matrix organization**.

(same as E4th Ch4 Q8)

如本章所述，项目经理的工作环境可能对其管理项目的能力产生重大影响。描述在以下组织结构中，可能有助于或阻碍项目经理工作的因素：职能型组织、学科型组织和矩阵型组织。

(1) Functional Organization

Helping Factors:

- **Ideal** for projects that require work only related to the **functional department**.
- **Suited** for management of **long-term projects** such as highways or electrical power lines.
- **Good communication** within the functional department, where people can **share common expertise**.

Hindering Factors:

- **Difficult** for projects that require **expertise outside** the functional department.
- Excessive focus on one **functional department** may **adversely affect** the entire organization.
- For **large or complex projects**, **coordination across functional departments** may be challenging.

职能型组织

有利因素:

- 适合仅需**职能部门**相关工作的项目。
- 适合管理**长期项目**，如公路或电力线路项目。
- 职能部门内部**沟通良好**，人员可以**共享共同的专业知识**。

不利因素:

- 对需要**职能部门以外专业知识**的项目来说，**难以胜任**。
- 过于关注一个**职能部门**，可能**对整个组织产生不利影响**。
- 对于**大型或复杂项目**来说，**跨职能部门协调**可能较为困难。

简化版:

Functional Organization

- **Helping Factors:** Consistent expertise within department, good communication, suited for long-term projects.
- **Hindering Factors:** Difficult cross-department coordination, limited expertise, impact on organization as a whole.

职能型组织

- **有利:** 部门内专业一致，沟通顺畅，适合长期项目。

不利: 跨部门难协调，专业局限，影响整体

(2) Discipline Organization

Helping Factors:

- **Technical specialists** are grouped together, enabling them to **share information and expertise**.
- Focus is on providing **technical expertise**, which tends to **enhance project quality**.
- Team members feel **comfortable and stable** as they work with colleagues with the same background, creating a sense of a **permanent home**.

Hindering Factors:

- Excessive focus on **technical expertise** may **distract** attention from **costs and schedules**.

- **Separation of disciplines** may lead to **competition and conflicts** between departments.
- As the number of **disciplines increases**, **coordination of complex projects** becomes more challenging.

学科型组织

有利因素: • 技术专家聚集在一起，便于共享信息和专业知识。 • 专注于提供技术专长，这通常可以提升项目质量。 • 团队成员感到舒适和稳定，因为他们与具有相同背景的同事一起工作，有一种长期归属感。	不利因素: • 过度关注技术专长，可能会忽视成本和进度。 • 学科间的分离可能导致部门之间的竞争和冲突。 • 随着学科数量增加，协调复杂项目变得更加困难。
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简化版:

Discipline Organization

- **Helping Factors:** Specialists gathered, quality improvement, strong sense of belonging.
- **Hindering Factors:** Neglect of cost/schedule, departmental competition, complex projects hard to coordinate.

学科型组织 • 有利: 技术专家集中，提升质量，团队归属感强	不利: 忽视成本进度，部门竞争，复杂项目难协调
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(3) Matrix Organization

Helping Factors:

- **Focus** is on the **project**, rather than on the functional department or discipline groups.
- Emphasis on **integrating costs, schedules, and quality** for the overall success of the project.
- **Greater flexibility** in personnel use, while maintaining **one person with total accountability**.

Hindering Factors:

- **Strong department managers** of team members may **interfere** with the project manager's role.
- **Conflicts** can arise if a team member has more **seniority** than the project manager.
- **Dual loyalty** can create challenges, as team members may feel loyalty to both the **project** and their **home departments**.

矩阵型组织

有利因素: • 聚焦于项目本身，而非职能部门或学科组。 • 强调成本、进度和质量的整合，以确保项目成功。 • 在人员使用上具有更大的灵活性，同时维持一个人对项目的全部责任。	不利因素: • 团队成员所属部门的强势经理可能会干扰项目经理的角色。 • 当团队成员比项目经理资历更高时，可能会引发冲突。 • 团队成员可能会产生双重忠诚，既忠于项目，又忠于所属部门，因此较难处理。
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简化版:

Matrix Organization

- **Helping Factors:** Project-focused, flexible staffing, integrates cost, schedule, and quality.
- **Hindering Factors:** Departmental interference, seniority conflicts, dual loyalty challenges.

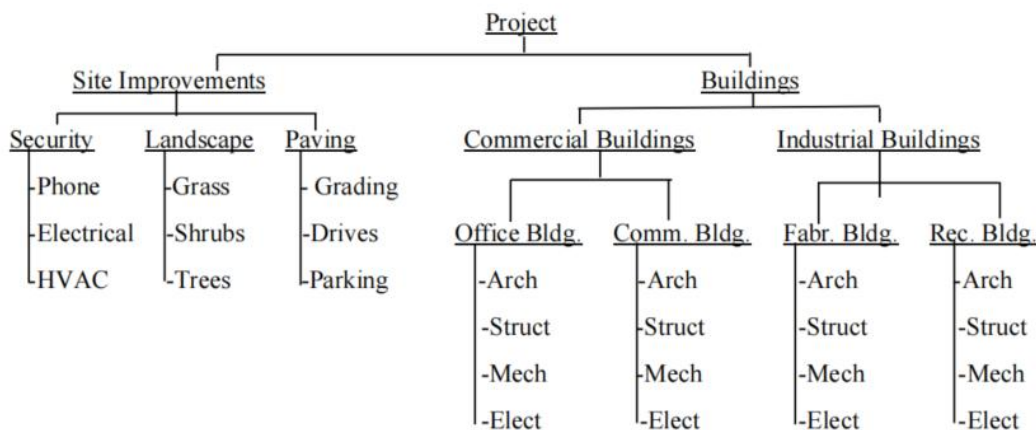
学科型组织 • 有利: 聚焦项目，灵活用人，综合成本进度质量	不利: 部门干预，资历冲突，双重忠诚难处理
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2. You are the **project manager** of a project that has the following facilities: **security entrance, driveways, parking, landscape, small office building, fabrication building, communication building, and a recreational building**. Develop a **preliminary work breakdown structure** for the project. Identify the **engineering disciplines** required for the design of the project.
(same as E4th Ch3 Q1)

你是一个项目的项目经理，该项目包含以下设施：安保入口、车道、停车场、景观、小型办公楼、制造楼、通信楼和娱乐楼。为该项目制定一个初步的工作分解结构，并识别项目设计所需的工程学科。

There are numerous configurations of a WBS for any project, depending on the **project team** and how they choose to **divide the project into identifiable work**. The **WBS will vary**, depending on the **work** to be performed by **in-house personnel** and the work that will be **contracted outside**. What is important is to **include all items** that are required to produce the **project**. The following **abbreviated WBS** is provided for this project. **Students** should review **Appendix A** as a guide for a more **detailed illustrative WBS** of a similar type of project.

任何项目的 **WBS**（工作分解结构）都可以有多种配置，具体取决于项目团队以及他们选择如何将项目划分为可识别的工作。**WBS** 的内容会有所不同，取决于由内部人员完成的工作和将外包的工作。关键在于包含所有完成项目所需的项。以下是本项目的简略版 **WBS**。学生应参阅附录 A，以获取类似项目的详细示例 **WBS** 作为参考。



简写	全称	中文翻译	简写	全称	中文翻译
Proj.	Project	项目	Buildings	Buildings	建筑物
Site Improvements	Site Improvements	场地改进	Commercial Buildings	Commercial Buildings	商业建筑
Security	Security	安保	Industrial Buildings	Industrial Buildings	工业建筑
Phone	Telephone Systems	电话系统	Office Bldg.	Office Building	办公楼
Electrical	Electrical Systems	电气系统	Comm. Bldg.	Communication Building	通信楼
Landscape	Landscape	景观	Fabr. Bldg.	Fabrication Building	制造楼
Grass	Grass	草地	Rec. Bldg.	Recreational Building	娱乐楼
Shrubs	Shrubs	灌木	Arch	Architecture	建筑设计
Trees	Trees	树木	Struct	Structural Engineering	结构工程
Paving	Paving	铺路	Mech	Mechanical Engineering	机械工程
Grading	Grading	场地平整	Elect	Electrical Engineering	电气工程
Drives	Driveways	车道	HVAC	Heating, Ventilation, and Air Conditioning	供暖、通风和空调
Parking	Parking	停车场			

3. Describe the **purpose of a kick-off meeting** for a project. What are **typical items** that are presented and discussed during the meeting, including **specific assignments** that a project manager should ask of **team members**?
(same as E4th Ch3 Q2)

描述项目启动会议的目的。在会议期间，通常会介绍和讨论哪些内容，包括项目经理应向团队成员分配的具体任务。

- The **kick-off meeting** provides an **orientation** for **team members**, including **goals, objectives, needs, requirements**, and the **general scope of work**.
 - Items to present include the **project title, account number**, and **identification of areas of expertise** and the **responsibility** of team members.
 - The **project manager** reviews the **preliminary work plan**, and how to handle **design, procurement, and construction**.
 - Items to discuss include **priorities, needs, guidelines, limitations, problems**, the **approved budget**, and the **required completion date**.
 - Ask **team members** to review the **scope of work** in their area of expertise, develop a **preliminary schedule**, and prepare a **preliminary estimate** for the cost of their work.
 - Ask team members to **identify problems** and develop a **design work package** for their work within **two weeks**.
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- 启动会议为团队成员提供导向，包括目标、宗旨、需求、要求以及工作的一般范围。
 - 需要介绍的内容包括项目名称、账号、专业领域的识别以及团队成员的职责。
 - 项目经理回顾初步工作计划，以及如何处理设计、采购和施工。
 - 需要讨论的内容包括优先事项、需求、指导方针、限制、问题、批准的预算和要求的完成日期。
 - 要求团队成员审查他们专业领域的工作范围，制定初步进度表，并准备其工作的初步成本估算。
 - 要求团队成员识别问题并在两周内为他们的工作制定设计工作包。

简化版：

Purpose of Kick-off Meeting: Provide orientation, clarify goals and requirements.

Items to Present: Project title, account number, responsibilities, areas of expertise.

Project Manager Reviews: Work plan, handling of design, procurement, and construction.

Discussion Topics: Priorities, needs, budget, completion date.

Assignments:

- Review scope of work
- Develop preliminary schedule
- Prepare preliminary cost estimate
- Identify issues, complete design work package within 2 weeks

项目启动会议目的：为团队成员提供导向，明确目标 and 需求。

介绍内容：项目名称、账号、职责、专业领域。

项目经理回顾：工作计划、设计、采购、施工处理方式。

讨论事项：优先事项、需求、预算、完成日期。

任务分配：

- 审查工作范围
- 制定初步进度表
- 初步成本估算
- 识别问题，2周内完成设计工作包

4. Briefly describe the **contents of a design work package**, who assigns it, when it is assigned, who completes it, and when it should be completed.
(same as E4th Ch3 Q3)

简要描述设计工作包的内容、由谁分配、何时分配、谁来完成以及何时完成。

- The **work package** is a **detailed description** of the **work to be performed**, including the **cost** and **schedule** for completing the work.
 - **Work packages** are **assigned by the project manager** at the **kick-off meeting**, after the **overall requirements** of the project are known.
 - Each **team member** is responsible for **preparing a work package** for the work that he or she will be performing in the project.
 - **Work packages** of team members should be completed within **two weeks** of the **kick-off meeting** so the **project manager** can integrate them into a **total project work plan**.
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- 工作包是对要完成的工作的**详细描述**，包括完成该工作的**成本和进度**。
 - 工作包由项目经理在启动会议上分配，前提是已了解项目的**总体要求**。
 - 每个**团队成员**负责为其将在项目中执行的工作**准备一个工作包**。
 - 团队成员的工作包应在启动会议后的**两周内**完成，以便项目经理将其整合到**整体项目工作计划**中。

简化版：

Contents of Design Work Package: Detailed description of work, cost, schedule. Assigned by: Project manager. Assigned at: Kick-off meeting. Completed by: Team members. Completion Time: Within two weeks after the kick-off meeting.	设计工作包内容： 详细描述工作、成本、进度。 分配者： 项目经理。 分配时间： 启动会议。 执行者： 团队成员。 完成时间： 启动会议后两周内。
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5. As the **project manager** for a project, discuss **options** that might be considered for one of your team members who is having difficulty completing a **design work package**. Discuss options for each of the following situations: **difficulty in defining the scope of work**, **difficulty in finding experienced personnel** that can be assigned to tasks in the work package, and **difficulty in estimating the time** that it may take to complete the work.
(same as E4th Ch3 Q4)

作为项目的**项目经理**，讨论可能针对一位在完成**设计工作包**时遇到困难的团队成员考虑的**选项**。针对以下情况分别讨论选项：**定义工作范围的困难**、**找到可以分配给工作包任务的经验丰富人员的困难**以及**估算完成工作所需时间的困难**。

(1) Difficulty in Defining Scope:

- Discuss the **end product** that needs to be produced in their **area of expertise**.
- Prepare a list of **deliverables** (plans, specifications, tests, reports, etc.) that are needed.
- Ask the team member to **consult with others** to get help in their **area of expertise**.
- Review the work of **similar past projects** to understand the work performed to complete those projects.

定义工作范围的困难:

- 讨论需要在**其专业领域**中生产的**最终产品**。
- 准备**交付物**列表（计划、规格、测试、报告等）。
- 要求团队成员**咨询其他人**，以获得其专业领域的帮助。
- **回顾类似的过往项目**的工作，以了解完成这些项目所执行的工作。

简化版:

Difficulty in Defining Scope: • Clarify end product • List deliverables • Consult other experts • Refer to similar projects	定义工作范围的困难: • 明确最终产品 • 列出交付物 • 咨询其他专家 • 参考类似项目
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(2) Difficulty Finding Experienced Personnel:

- Suggest the team member **mentor and train** the person who lacks experience.
- Check for available **training** within the company and outside.
- Evaluate **potential consultants** who may be used on a contract basis to perform the work.
- Establish a **control system** to review the work of the inexperienced personnel.

找到有经验人员的困难:

- 建议团队成员**指导和培训**缺乏经验的人员。
- 查看公司内部和外部是否有可用的**培训资源**。
- 评估可能以合同形式完成工作的**潜在顾问**。
- 建立一个**控制系统**，以审核缺乏经验人员的工作。

简化版:

Difficulty Finding Experienced Personnel:

- **Mentor** less experienced team members
- Look for **training resources**
- Evaluate **external consultants**
- Set up a **control system**

找到有经验人员的困难:

- 指导培训缺乏经验的人员
- 查找培训资源
- 评估外部顾问
- 建立控制系统

(3) Difficulty in Estimating the Time:

- Break the **work package** into **smaller segments** to better identify the work.
- Prepare **range estimates** (optimistic and pessimistic time) for each work component.
- Consult with others who have done **similar work** to obtain their actual **expended time**.
- Estimate the time for **known work**, then add a reasonable **contingency** for unknowns.

估算时间的困难:

- 将工作包拆分成更小的部分，以更清晰地识别工作内容。
- 为每个工作部分准备范围估算（乐观时间和悲观时间）。
- 与从事过类似工作的其他人交流，以获得他们的实际工作时间。
- 先估算已知工作的时间，然后为未知部分添加合理的预备时间。

简化版:

Difficulty in Estimating Time:

- Break work into **smaller parts**
- Make **range estimates** (optimistic/pessimistic)
- Check **similar project times**
- Add contingency for **unknowns**

估算时间的困难:

- 将工作分解为小部分
- 做范围估算（乐观/悲观）
- 参考类似项目时间
- 为未知因素添加预备时间

6. You are the **project manager** for a project and have just compiled all the **design work packages** and found that the **costs exceed the approved budget**. Describe the **options** that you would consider to manage the **discrepancy**.
(same as E4th Ch3 Q5)

你是项目的**项目经理**，刚刚编制了所有**设计工作包**，发现**成本超出了批准的预算**。描述你会考虑的**选项**以管理**差异**。

- **Call a project team meeting** and inform members that **anticipated costs exceed the budget**.
 - Ask each **team member** to **recheck their budgets** to determine if an **error** has been made.
 - Ask team members to **assess the possibility of duplication** of work in **work packages**.
 - Ask each team member to provide the **amount of allowances and contingency** in their work packages.
 - **Evaluate the possibility** of using **outside contract work** for some parts of the project.
 - If none of the above succeeds, **inform your supervisor** to determine possible actions.
 - **Inform the marketing group** that secured the project, to prevent future problem projects.
 - **Final option**: call a meeting with the **owner** to explain the situation and resolve the problem.
 - **Negotiate** an increase in **budget** with the owner or **adjust the scope** to match the budget.
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- 召开项目团队会议，告知成员**预计成本超出预算**。
 - 要求每位**团队成员**重新检查他们的**预算**，以确定是否存在**错误**。
 - 要求团队成员**评估工作包中是否有重复工作的可能性**。
 - 要求每位团队成员提供其工作包中的**补贴和应急金额**。
 - **评估可能性**，考虑将项目的部分工作**外包**。
 - 如果上述方法均未奏效，**通知上级**以确定可能的解决方案。
 - 告知负责该项目的**市场部门**，以防止未来发生类似问题项目。
 - **最终选择**：与**业主**召开会议，解释情况并解决问题。
 - 与**业主**协商增加**预算**，或**调整项目范围**以符合**预算**。

简化版：

Options for Managing Budget Overrun:

- Hold a **project team meeting** to inform about the overrun
- **Check budgets** for errors
- **Assess duplication** of work
- Gather information on **allowances and contingencies**
- Consider **outsourcing parts** of the project
- **Notify supervisor**
- Inform the **marketing department** to prevent future issues
- Final option: **meet with the owner**
- **Negotiate budget increase** or **adjust project scope**

应对预算超支的选项：

- 召开项目团队会议，告知超支情况
- **检查预算**，寻找错误
- 评估重复工作的可能性
- 获取**补贴和应急金额**信息
- 考虑**外包部分工作**
- **通知上级**
- 告知**市场部门**，防止类似问题
- **最终选择**：与**业主**会议
- **协商增加预算**或**调整项目范围**

7. The basic contents of a **project work plan** are shown in **Table 6-3**. Identify the contents that relate to the fundamental questions of **Who?**, **Does what?**, **When?**, and **How much?** (similar as E4th Ch3 Q6)

项目工作计划的基本内容显示在表 6-3 中。识别与以下基本问题相关的内容：谁？做什么？什么时候？花费多少？

E4th Ch3Q6:

The basic contents of a **project work plan** are shown in **Table 3-4**. Identify the contents that relate to the fundamental questions of **Who? Does What? When? How Much? What can go wrong? And What is accomplished?**

- **Project Organization Chart** - refers to who will be doing the work.
 - **General Description, Work Breakdown Structure, Detailed Task List & Work Packages** - refers to what work will be done.
 - **Major Milestone Summary, Basis of Schedule, Project Schedule** (contractor's and designer's schedules) - refers to when the work will be done.
 - **Basis of Estimate, Project Budget, Project Cost Estimate, Cost/Resource Loaded Schedule** - refers to how much it will cost to do the work.
 - **Risk Matrix, Risk Assessment, Impacts, and Mitigations** - refers to what can go wrong.
 - **Design and Construction Progress Measurement, Procurement Progress Measurement, Change Logs** - refers to what is accomplished.
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- 项目组织结构图 - 表示谁将执行工作。
 - 一般描述、工作分解结构、详细任务列表和工作包 - 表示将要完成的具体工作。
 - 主要里程碑总结、进度基础、项目进度表（包括承包商和设计方的进度）- 表示何时完成工作。
 - 估算基础、项目预算、项目成本估算、成本/资源负载进度表 - 表示完成工作需要多少成本。
 - 风险矩阵、风险评估、影响及缓解措施 - 表示可能出错的事项。
 - 设计和施工进度测量、采购进度测量、变更日志 - 表示已完成的成果。

简化版:

Basic Contents of a Project Work Plan:	项目工作计划基本内容简化版:
• Who: Project Organization Chart • Does What: Work Description, Work Breakdown Structure, Task List, Work Packages • When: Milestones, Schedule Basis, Project Schedule • How Much: Basis of Estimate, Project Budget, Cost Estimate, Resource Schedule • What Can Go Wrong: Risk Matrix, Risk Assessment, Impacts & Mitigations • What Is Accomplished: Progress Measurement, Procurement Measurement, Change Logs	• 谁: 项目组织结构图 • 做什么: 工作描述、工作分解结构、任务列表、工作包 • 何时: 里程碑、进度基础、项目进度表 • 多少成本: 估算基础、项目预算、成本估算、资源进度表 • 可能出错: 风险矩阵、风险评估、影响与缓解措施 • 完成了什么: 进度测量、采购测量、变更日志

Directory	←	Who?
Project title and number		
Project objectives and scope		
Project organizational chart		
Tasks	←	Does what?
Detailed listing of tasks		
Grouping of tasks		
Work packages		
Schedule	←	When?
Sequencing and interdependencies of tasks		
Anticipated duration of each task		
Calendar start and finish dates of tasks		
Budget	←	How much?
Labor-hours and cost of staff for each task		
Other expenses anticipated for each task		
Billing approach and anticipated revenue by month		
Risk Assessment	←	What can go wrong?
List of risks and their probability of occurrence		
Impact of each risk on quality, cost, and schedule		
Strategy to prevent and mitigate each risk		
Measurement	←	What is accomplished?
Accomplishment of tasks		
Completion of work packages		
Number of drawings produced		

Project Plan of Work	
Project Plan Elements	
General Information	
Project title and number Project objectives and scope Project organizational chart Project contact list	Who ...
Project Objective	
Objective general description Specific objectives Justification general description	
Project Scope	
General description Work breakdown structure Task groups Detailed task list Work packages Scope change log	Does what ...
Project Costs	
Basis of estimate Budget summary Budget—labor-hours and cost of staff for each task Estimate Engineer's construction estimate Contractor's construction bid (schedule of values) Other expenses anticipated for each task Billing approach and anticipated revenue or expenditure by month	For how much ...
Project Schedule	
Basis of schedule Major milestone summary Project schedule showing sequencing, interdependencies, and estimated duration of tasks Calendar showing start and finish dates for tasks Scheduling threats, project schedule risks and opportunities Engineer's construction schedule Contractor's construction schedule	When ...
Risk Management Plan	
Risk matrix Risk assessment including list of risks, probability of occurrence, and expected impacts Impact of each risk on quality, cost, schedule Risk mitigation plan	What can go wrong ...
Monitor & Control Plan	
Design progress measurement methods Construction progress measurement methods Procurement progress measurement methods Change (budget, schedule baselines) procedures	What is accomplished ...

8. For the **basic components of a project work plan** shown in **Table 6-3**, identify the components that relate to the **organization structure**, **work breakdown structure**, **cost breakdown structure**, and the **Critical Path Method** for project scheduling.

对于显示在表 6-3 中的项目工作计划的基本组成部分，识别与组织结构、工作分解结构、成本分解结构以及用于项目排程的关键路径法相关的组成部分。

(1) Organizational Structure

- **Project Organizational Chart:** This component shows the structure of the project team, defining roles and responsibilities, which identifies **who** will perform the work. It is directly related to the Organizational Breakdown Structure (OBS).
- **Project Contact List:** Provides contact information for team members, supporting communication and organizational management within the project.

组织结构（Organizational Structure）

- **项目组织结构图（Project Organizational Chart）：**此部分显示了项目团队的组成、各成员的角色和职责，定义了**谁**将执行工作，确保项目的组织清晰。与组织分解结构（OBS）直接相关。
- **项目联系清单（Project Contact List）：**提供了团队成员的联系信息，支持项目的沟通和组织管理。

(2) Work Breakdown Structure (WBS)

- **Work Breakdown Structure (WBS):** The core of the project plan, used to break down the project into manageable tasks, clarifying **what work needs to be done**. It serves as the foundation for all other breakdown structures.
- **Detailed Task List:** Based on the WBS, this breaks tasks down further into specific work packages.
- **Work Packages:** Specifies the content, responsible person, and deliverables for each task, closely related to the WBS.

工作分解结构（Work Breakdown Structure, WBS）

- **工作分解结构（Work Breakdown Structure）：**这是项目计划的核心，用于将项目划分为更小的可管理任务，明确**需要完成的工作**，是所有其他分解结构的基础。
- **详细任务列表（Detailed Task List）：**基于 WBS，将任务分解为更具体的工作包。
- **工作包（Work Packages）：**具体列出了每个工作项的内容、责任人和输出成果，与 WBS 紧密关联。

(3) Cost Breakdown Structure (CBS)

- **Budget Summary:** Includes the main project costs, directly related to the CBS.
- **Cost Breakdown Structure (CBS):** Contains the cost estimates for each project component and uses an account code system to track expenses, essential for **managing project costs**.
- **Budget—Labor-Hours and Cost of Staff for Each Task:** Details the costs for each task, helping the project manager monitor budget and spending.

成本分解结构（Cost Breakdown Structure, CBS）

- **预算总结（Budget Summary）：**包括所有主要项目成本，与 CBS 密切相关。
- **成本分解结构（CBS）：**包括对项目各个部分的成本估算，并通过账户编码体系进行追踪，用于管理项目成本。
- **每项任务的工时和人员成本预算（Budget—labor-hours and cost of staff for each task）：**细化了各任务的成本，帮助项目经理监控预算和成本的使用情况。

(4) Critical Path Method (CPM)

- **Project Schedule:** By scheduling each task, the project manager ensures the project stays on track. The CPM can identify the project's critical path, enabling effective schedule management.
- **Major Milestone Summary:** Helps establish key project checkpoints, working with the CPM to maintain control over the project timeline.
- **Calendar Showing Start and Finish Dates for Tasks:** Provides timing for each task, allowing the project manager to use CPM for tracking and adjusting the schedule.

关键路径法（Critical Path Method, CPM）

- **项目进度表（Project Schedule）：**通过对各任务进行排程，确保项目按时完成。CPM 可以用于识别项目中的关键路径，从而有效管理进度。
- **主要里程碑总结（Major Milestone Summary）：**帮助确定项目的关键节点，与 CPM 结合，确保项目进度的整体控制。
- **日历（任务的开始和结束日期）（Calendar showing start and finish dates for tasks）：**提供了每个任务的时间安排，使项目经理可以通过 CPM 监控和调整进度。

简化版：

(1) Organizational Structure • Project Organizational Chart: Defines roles and responsibilities • Project Contact List: Supports communication and management (2) Work Breakdown Structure (WBS) • Work Breakdown Structure (WBS): Breaks down project tasks • Detailed Task List: Further specifies tasks • Work Packages: Lists content, responsible person, deliverables (3) Cost Breakdown Structure (CBS) • Budget Summary: Main project costs • Cost Breakdown Structure (CBS): Cost estimates, tracks expenses • Labor-Hours and Cost Budget: Details costs for each task (4) Critical Path Method (CPM) • Project Schedule: Arranges task sequence, identifies critical path • Major Milestones: Sets key project checkpoints • Calendar (Start and Finish Dates): Provides task timing	(1) 组织结构（Organizational Structure） • 项目组织结构图：定义角色和职责 • 项目联系清单：支持沟通和管理 (2) 工作分解结构（Work Breakdown Structure, WBS） • 工作分解结构（WBS）：项目任务分解 • 详细任务列表：进一步细化任务 • 工作包：列出内容、责任人、交付物 (3) 成本分解结构（Cost Breakdown Structure, CBS） • 预算总结：主要项目成本 • 成本分解结构（CBS）：成本估算，追踪成本 • 任务工时和成本预算：各任务的成本细节 (4) 关键路径法（Critical Path Method, CPM） • 项目进度表：安排任务进度，识别关键路径 • 主要里程碑：设定项目关键节点 • 日历（开始和结束日期）：提供任务时间安排
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9. The development of a **work plan** for a project follows **four phases: project definition, framework, schedule, and tracking**. Identify and briefly describe the parts that make up each of these four phases. (similar as E4th Ch3 Q7)

项目的工作计划制定包含四个阶段：项目定义、框架、进度和跟踪。识别并简要描述构成这四个阶段的各部分。

E4th Ch3Q7:

The development of a **work plan** for a project follows **four phases: project definition, framework, schedule, and tracking**. Identify and briefly describe the parts that make up each of these four phases and **how they are addressed in the PEP**. (PEP = Project Execution Plan)

(1) Project Definition:

- **Owner's feasibility study** includes **economic or benefit/cost analysis** and the features the project should have to be usable by the owner.
- The **project manager's initial review** of all **backup material** prepared by the owner and/or the group that presented and obtained **management approval** of the project.
- The **initial meeting** between the **project manager** and the **owner** to discuss the project in general and to get answers to any questions the project manager might have before starting work.
- **Joint agreement** between the **owner** and **project manager** to confirm the **scope, budget, and schedule** for the project and the **level of quality** expected.

项目定义：

- 业主的可行性研究，包括经济或成本/收益分析，以及项目应具备的功能，以便业主使用。
- 项目经理对业主和/或提出并获得项目管理批准的团队所准备的所有备选材料的初步审核。
- 项目经理与业主的首次会议，讨论项目概况，并解答项目经理在开工前可能有的任何问题。
- 业主和项目经理之间的共同协议，以确认项目的范围、预算、进度，以及预期的质量水平。

简化版：

Four Phases of a Project Work Plan

Project Definition:

- Feasibility study: cost/benefit analysis, functional requirements
- Initial review: project backup materials
- Initial meeting: address questions, discuss project overview
- Joint agreement: confirm scope, budget, schedule, quality

项目工作计划的四个阶段

项目定义：

- 可行性研究：成本/收益分析，功能要求
- 初步审核：项目备选材料
- 首次会议：解答问题，讨论项目概况
- 共同协议：确认范围、预算、进度、质量

(2) Project Framework:

- **Project manager** develops the initial **work breakdown structure** and works with **project team members** to expand it in more detail.
- The **organizational breakdown structure** of the project manager's organization is merged with the **work breakdown structure** to define **who is doing what**.
- The **cost breakdown structure** is developed, containing the **code of accounts** for the project, based on the work items in the **work breakdown structure**.
- The **project manager** and **discipline managers** work together to **select team members**.

项目框架:

- 项目经理制定初步的工作分解结构，并与项目团队成员合作，进一步细化。
- 项目经理组织的组织分解结构与工作分解结构合并，以定义谁负责什么工作。
- 制定成本分解结构，其中包含项目的科目编码，基于工作分解结构中的工作项。
- 项目经理和学科经理共同工作以选择团队成员。

简化版:

Project Framework: • Develop Work Breakdown Structure (WBS) • Merge Organizational Breakdown Structure with WBS: clarify responsibilities • Cost Breakdown Structure: code of accounts • Select team members	项目框架: • 制定工作分解结构（WBS） • 合并组织分解结构与 WBS: 明确职责 • 成本分解结构: 科目编码 • 选定团队成员
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(3) Project Scheduling:

- **Tasks** in the **work breakdown structure** are sequenced into an overall **project schedule**.
- The **schedule** is **resource-loaded**, and reports are generated to **forecast the project staffing needs**.
- A **cost analysis** is performed to show the **planned costs** throughout the **project duration**.

项目进度安排:

- 将工作分解结构中的任务按顺序安排成总体项目进度表。
- 进度表加载资源，并生成报告以预测项目的人力需求。
- 进行成本分析，以显示整个项目期间的计划成本。

简化版:

Project Tracking: • Tracking system: work, cost, schedule • Periodic reports: progress on work, cost, and time • Trend analysis: compare planned vs. actual, forecast costs and completion date	项目进度安排: • 排列任务: 形成整体进度表 • 资源分配: 生成人员需求预测 • 成本分析: 显示项目期间的计划成本
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(4) Project Tracking:

- A **system** is developed to track the three components of a project: **the work to be performed**, the **costs expended**, and the **schedule for all work activities**.
- **Periodic reports** are compiled to show **progress** with respect to **work, cost, and time**.
- An **analysis** of the planned and actual **costs**, planned and actual **work accomplished**, and planned and actual **schedule** is conducted to develop **trends** in the project.
- Analysis of **trends** in the project to **forecast the cost at completion** and **forecast the finish date** of the project.

项目跟踪:

- 开发一个**系统**来跟踪项目的三个组成部分:**要完成的工作、支出的成本和所有工作活动的进度**。
- 编制定期报告, 以显示**工作、成本和时间**方面的进展。
- 分析计划和实际的**成本**、计划和实际的**完成工作**以及计划和实际的**进度**, 以确定项目中的**趋势**。
- 分析项目的**趋势**, 以**预测项目完成时的成本**以及项目的**完成日期**。

简化版:

Project Scheduling:

- Sequence tasks: form overall schedule
- Resource loading: forecast staffing needs
- Cost analysis: planned costs over project duration

项目跟踪:

- 跟踪系统: 工作、成本、进度
- 定期报告: 工作、成本、时间进展
- 分析趋势: 计划与实际比较, 预测成本和完成日期

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Figure 6-11 Phases of development of workplan.

